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IS MULTIMODAL SPECULATIVE DECODING READY FOR DIFFUSION-BASED PARALLEL DRAFTING? A SURVEY AND EMPIRICAL DIAGNOSIS

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ABSTRACT

Speculative decoding accelerates autoregressive generation by allowing a lightweight drafter to propose future tokens while a target model verifies them in parallel. Its lossless guarantee has motivated a line of work that pushes the drafter itself toward parallel generation—the most recent prominent milestone is block-parallel generative approaches, including diffusion-based drafting methods such as DFlash and DSpark, achieving a 3.6× speedup on common daily conversation tasks. While diffusion-based block-parallel drafting has been explored in text-only LLMs, its transferability to multimodal models remains an open question. Existing multimodal speculative decoding efforts focus on input compression, adapter alignment, candidate coverage, or modality-specific verification; however, block-parallel generative drafting remains largely unexplored. To bridge this gap, this paper combines a modality-centered survey with a cross-architecture empirical study to ask: Is multimodal speculative decoding ready for diffusion-based parallel drafting? In this survey, we systematically analyze a wide spectrum of multimodal models—spanning Vision-Language, Video-Language, Audio, and Vision-Language-Action (VLA) architectures—from the dual perspectives of drafting parallelism and cross-modal information interaction. We introduce a unified taxonomy that isolates drafter-side parallelism from orthogonal design choices such as tree construction and verification strategies. Furthermore, we provide a comprehensive empirical evaluation of existing methods under varying degrees of parallelism across standardized multimodal benchmarks, such as OCR, VQA, visual reasoning, and image captioning. Finally, we summarize the limitations of current approaches, discuss open challenges, and outline promising future directions for this rapidly evolving field.

1 INTRODUCTION

Multimodal generative models have achieved remarkable progress in a wide range of applications, including visual question answering, document understanding, chart reasoning, video analysis, speech interaction, and embodied control. Despite these advances, inference remains dominated by autoregressive decoding. After multimodal context encoding and prefill, the decoder still generates tokens sequentially, requiring one expensive target model execution for each output token. This limitation becomes increasingly severe for long-form reasoning, OCR-heavy generation, structured documents, and interactive multimodal agents.

Speculative decoding reduces this autoregressive bottleneck by introducing a lightweight drafter that proposes future tokens, which are then verified by the target model in parallel (Leviathan et al., 2023; Chen et al., 2023). The achievable acceleration depends on two complementary factors: how

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many draft tokens can be accepted by the target model and how efficiently the draft candidates can be generated. Existing speculative decoding methods have therefore evolved along two directions.

The first direction improves verification efficiency by increasing candidate coverage. Tree-based speculative decoding methods, including SpecInfer (Miao et al., 2024), Sequoia (Chen et al., 2024), DDTree (Ringel & Romano, 2026), and TAPS (Wang et al., 2026a), organize multiple candidate continuations into tree structures and verify them in parallel, improving the expected accepted tokens under a fixed verification budget. These approaches mainly optimize the utilization of each target model forward pass while retaining sequential or weakly parallel drafting.

The second direction improves drafter-side parallelism. Traditional autoregressive drafters generate a block of K future tokens through K serial draft steps. Feature-based approaches such as EAGLE (Li et al., 2024a;b; 2025b), multi-head prediction methods such as Medusa (Cai et al., 2024), and multi-token prediction (Gloeckle et al., 2024) reduce drafting latency by predicting multiple future positions in parallel. More recently, diffusion-based block-parallel drafting has emerged as a stronger form of output-side parallelism. DFlash (Chen et al., 2026) formulates future-token generation as a block-level diffusion process, while DSpark (Cheng et al., 2026) introduces lightweight intra-block dependency modeling and confidence-aware scheduling. These methods shift speculative decoding from token-wise prediction toward jointly generating future blocks.

Together, these advances change the central question of speculative decoding from "Can the target verify multiple tokens simultaneously?" to "Can the drafter generate sufficiently accurate future blocks without paying the cost of serial drafting?"

Multimodal speculative decoding introduces additional challenges beyond text-only generation. Existing approaches mainly investigate how a drafter should access visual information, how redundant visual tokens should be compressed, how target representations can improve draft quality, and how verification should account for perceptual or functional equivalence (Gagrani et al., 2024; Ganesan et al., 2025; Hu et al., 2025b; Huang et al., 2025; Wang et al., 2025a). These studies demonstrate that multimodal conditioning is critical for maintaining draft–target agreement. However, most existing methods remain based on token-wise or semi-parallel drafting. Whether block-parallel drafting can provide practical acceleration for frozen, off-the-shelf multimodal targets under exact verification remains largely unexplored.

To analyze this question systematically, we introduce a three-level taxonomy based on drafter-side parallelism. L0 advances one future token per forward step, covering autoregressive drafters and feature-based approaches such as EAGLE. L1 predicts multiple predefined future positions from a shared prefix, including methods such as Medusa and multi-token prediction. L2 treats the future block as a jointly generated or refined unit, including recent block-parallel methods such as DFlash and DSpark. Candidate organization strategies, such as tree branching and verification rules, are orthogonal to this taxonomy. Under this view, most existing multimodal speculative decoding methods operate at L0 or L1. Existing L2 multimodal systems either modify the target architecture for self-speculation (Wu et al., 2026; Fu et al., 2026) or relax exact verification (Hu & Zhang, 2026; Niu et al., 2026). The general applicability of L2 drafting for frozen multimodal targets therefore remains an open question.

We investigate this question through a controlled empirical study of diffusion-based parallel drafting on multimodal models. Interestingly, directly transferring an L2 drafter trained on Qwen3 leads to limited draft–target agreement, while the same block-parallel drafting paradigm achieves substantially stronger alignment on newer multimodal variants. This observation suggests that multimodality itself is not the fundamental obstacle; instead, the effectiveness of L2 drafting depends on how multimodal information is represented, aligned, and utilized during generation. Therefore, rather than asking whether L2 drafting works for multimodal models, we study under what model, task, and system conditions block-parallel drafting can translate draft quality improvements into practical acceleration.

In this paper, we present a systematic empirical study of multimodal L2 speculative decoding readiness. Our contributions are:

1. **A unified taxonomy and empirical study of multimodal L2 drafting.** We establish an L0–L2 taxonomy based on drafter-side parallelism and evaluate block-parallel drafting across representative multimodal models with different scales and architectures.

2. **Analysis of multimodal conditioning bottlenecks.** We investigate how visual information affects draft–target agreement and end-to-end efficiency, revealing the trade-off between multimodal awareness and additional conditioning overhead.

3. **A readiness characterization for multimodal block-parallel decoding.** We analyze the impact of model characteristics, task and input conditions, and framework support on practical L2 acceleration, and summarize remaining challenges toward broader multimodal speculative decoding deployment.

The remainder of this paper is organized as follows. Section 2 introduces multimodal speculative decoding and the information-preservation requirements behind draft verification. Section 3 presents our L0–L2 taxonomy of drafter-side parallelism. Section 4 evaluates the readiness of block-parallel drafting for multimodal targets through cross-model evaluation, conditioning analysis, and framework support analysis. Section 5 synthesizes the empirical findings and discusses future directions before concluding in Section 6.

2 Multimodal Speculative Decoding

2.1 Foundations

Standard speculative decoding. Speculative decoding accelerates autoregressive generation with a draft-and-verify loop. It uses a lightweight drafter to propose multiple future tokens, allowing the target model to verify them in a single forward pass instead of decoding them one by one. Given a prefix $x_{<t}$, the drafter proposes K future tokens, and the target scores the drafted positions in parallel while also producing the next-token distribution after the draft block. Under exact speculative sampling, for $i = 0, \ldots, K-1$, a drafted token $\tilde{x}_{t+i}$ is accepted with probability

$$a_i(\tilde{x}_{t+i}) = \min\left\{1, \frac{p(\tilde{x}_{t+i} \mid x_{<t}, \tilde{x}_{t:t+i-1})}{q(\tilde{x}_{t+i} \mid x_{<t}, \tilde{x}_{t:t+i-1})}\right\}, \tag{1}$$

where $\tilde{x}_{t:t-1}$ is empty for $i = 0$. At the first rejection, the accepted prefix is retained and a correction token is sampled from the normalized positive residual between the corresponding target and draft distributions; after a full acceptance, the target contributes one additional token. This procedure preserves the target sampling distribution (Leviathan et al., 2023; Chen et al., 2023). Its benefit depends on whether the accepted output amortizes the cost of drafting and verification. Moreover, although the target evaluates multiple drafted positions in parallel, a conventional drafter produces them in causal order, so draft generation can remain a serial bottleneck.

Sequential and parallel drafting. Parallel drafting reduces this draft-side dependence rather than merely parallelizing target verification. It can take different forms, from predicting multiple future positions in one pass to generating or refining a future block as a coupled state (Stern et al., 2018; Cai et al., 2024; Gloeckle et al., 2024). DFlash exemplifies diffusion-based parallel drafting by using a lightweight block-diffusion model to generate a future block for verification by an autoregressive target (Chen et al., 2026). Section 3 formalizes the progression from sequential to increasingly parallel draft generation. These mechanisms are commonly studied for text generation, but multimodal settings introduce additional constraints on both input processing and output verification.

Multimodal speculative decoding. Unlike text-only SD, a multimodal drafter may need to process image, video, audio, or environment information that also conditions the target. Access to this information can improve target–draft agreement, but introduces additional modality-specific computation and memory cost. Some methods instead reuse target-derived features or draft without directly processing the multimodal input. Multimodal applications may also generate actions, acoustic or codec tokens, and visual tokens or latents rather than text tokens. For these outputs, exact token identity is not always the only useful notion of validity, and some methods adopt task- or representation-specific acceptance rules. The next subsection introduces the analytical dimensions used to compare these design choices across application domains.

2.2 Multimodal Speculative Decoding Landscape

We organize the multimodal speculative decoding literature considered in this survey into five application domains: vision–language models, video–language models, vision–language–action models, speech and audio models, and autoregressive visual generation. These domains differ in the multimodal information available during drafting, the outputs being generated, and their task and deployment constraints. Such differences affect both how speculative candidates are produced and how they can be verified, motivating application domain as the primary organization of the review.

Within each domain, we examine how candidate generation varies in its serial dependence across future positions. We annotate this property using the L0–L2 scale, which is formalized in Section 3. Methods with the same P-level can still differ substantially in how the drafter accesses multimodal information, how candidates are organized or obtained, how they are accepted or verified, and whether speculative computation adapts to the current input or state. We track these properties separately because they can change the effectiveness of speculative inference without changing future-position dependence. Table 1 summarizes these method-level characteristics, while the discussion below synthesizes the main design patterns and available evidence within each application domain.

Vision–language models. In VLMs, visual access can improve target–draft alignment, but its processing cost can offset the latency saved by accepting more draft tokens. Early work therefore asks whether the drafter needs direct visual input at all. SPD-MLLM shows that a language-only drafter can already accelerate a visually conditioned target, while adding a compact image adaptor produces task-dependent rather than uniform gains (Gagrani et al., 2024). MSD instead develops an explicitly multimodal drafter: it processes text and visual tokens separately and uses text-only followed by multimodal training to acquire both language-modeling and visual-perception capabilities (Lin et al., 2025a). Both retain an autoregressive draft trajectory, but establish the two endpoints between condition-blind and explicitly multimodal drafting.

Subsequent methods seek less redundant or more selective access to visual information. MASSV reuses target vision-encoder outputs through a lightweight projector (Ganesan et al., 2025); Huang et al.'s SpecVLM employs an elastic visual compressor (Huang et al., 2025); and ViSpec combines a compact vision adaptor with persistent global-feature injection (Kang et al., 2025). Other methods increasingly reuse representations already computed by the target. At inference time, DREAM cross-attends to cached target features while compressing the visual context; during training, it uses entropy-guided selection of intermediate target representations (Hu et al., 2025b). HiViS removes visual tokens from the drafter input and transfers visual semantics through target-derived hidden representations (Xie et al., 2026). TIGER makes this access step-dependent by routing visual tokens according to the current textual state and trains the drafter with a verifier-derived accepted-prefix reward (Vo et al., 2026). This progression moves from duplicating explicit visual inputs toward compressed, reused, and selectively exposed visual information.

Candidate topology and runtime control provide complementary improvements without necessarily changing future-position dependence. ViSpec also uses a dynamic draft tree, while Spec-LLaVA expands and prunes tree branches according to draft confidence (Huo et al., 2025) and SAGE adapts tree depth and width using online entropy (Tong et al., 2026). ViSkip instead uses vision-aware signals to adapt when speculative decoding is attempted (Shen et al., 2026). Trees broaden the candidate space, whereas ViSkip controls the allocation of speculative computation; neither mechanism alone removes the serial dependence of the underlying drafter.

SpecFLASH changes candidate generation more directly. Its semi-autoregressive drafter processes K placeholder tokens and predicts the next K token distributions in one pass (Wang et al., 2025b). We classify it as L1 because multiple fixed future offsets are predicted in parallel from a shared accepted context, rather than generated through a jointly evolving block state.

Block-parallel VLMs provide direct L2 examples. Fast-dVLM converts an autoregressive VLM into a KV-cache-compatible block-diffusion model and supports self-speculative block decoding (Wu et al., 2026). Nemotron-Labs-Diffusion jointly trains autoregressive and diffusion objectives within a shared architecture; its VLM variants can draft through diffusion and verify autoregressively (Fu et al., 2026). In both cases, the future block becomes the draft-side generative object rather than a collection of independently attached offset predictions.

Table 1: Taxonomy of representative multimodal speculative decoding methods across application domains, drafter designs, and speculative mechanisms.

Method	Drafter		Mechanism		
	L	Condition	Candidate form	Acceptance	Adaptation
Vision–Language Models					
SPD-MLLM (2024)	L0	Text / direct	Chain	Exact	Static
MASSV (2025)	L0	Target feat.	Chain	Exact	Static
DREAM (2025b)	L0	Target feat.	Chain	Exact	Static
HiViS (2026)	L0	Target feat.	Chain	Exact	Static
SpecVLM (image) (2025)	L0	Direct / target feat.	Chain / tree	Exact	Condition
ViSpec (2025)	L0	Direct	Tree	Exact	Candidate
TIGER (2026)	L0	Direct	Chain	Exact	Condition
Spec-LLaVA / SAGE (2025; 2026)	L0	Direct	Tree	Exact	Candidate
ViSkip (2026)	–	Target feat.	Inherited	Inherited	Execution
SpecFLASH (2025b)	L1	Direct / target feat.	Parallel offsets	Exact	Static
Fast-dVLM (2026)	L2	Direct	Joint block	Exact (greedy)	Static
Nemotron-Labs-Diffusion-VLM (2026)	L2	Direct	Joint block	Exact (greedy)	Static
Video–Language Models					
Sparse-to-Dense (2025)	L0	Direct	Chain	Exact	Candidate
SpecVLM (video) (2025)	L0	Direct / target feat.	Tree	Exact	Condition
Sparrow (2026b)	L0	Target feat.	Tree	Exact	Static
HIPPO / ParallelVLM (2026; 2026)	L0	Direct / target feat.	Chain	Exact	Condition / execution
LVSpec (2026)	L0	Direct	Chain	Relaxed (position)	Verification
Axis 1: Vision–Language–Action Models					
Spec-VLA (2025a)	L0	Direct	Chain	Relaxed (distance)	Static
HeiSD (2026b)	L0	Direct	Chain / retrieval	Relaxed (sequence)	Candidate
Reasoning-aware SD (diff.) (2026)	L2	Target feat.	Reasoning block	Exact (greedy)	Static
Realtime-VLA FLASH (2026)	L2	Direct	Action block	Consistency	Execution
Speech and Audio Models					
Speech SD (2025b)	L0	Text / direct	Chain	Relaxed (tolerance)	Static
SpecASR (2025)	L0	Direct	Tree	Exact	Candidate
ParaASR (2026)	L1	Direct	Parallel offsets	Exact	Static
VADUSA (2025a)	L1	Text / direct	Offsets / tree	Relaxed (tolerance)	Static
CTC-SSD (2026)	L1*	Direct	Frame hypothesis	Confidence-gated	Verification / execution
PCG (2026)	L0	Text / direct	Chain	Group-exact	Static
Autoregressive Visual Generation					
LANTERN (2025)	L0	Text / direct	Chain	Relaxed (latent)	Static
LANTERN++ (2025)	L0	Text / direct	Tree	Relaxed (latent)	Static
Spatial SD (2026)	L1	Text	Parallel rows	Correction-based	Static
SDVG (2026)	L2	Text	Video block	Quality-gated	Execution
Speculative Jacobi (2025b)	L2*	Text / direct	Iterative block	Convergence	Static

Notation. "Direct" denotes direct non-text conditioning; "target feat." denotes reused target features. P-level measures dependence across future draft positions, independent of candidate form: L0 = sequential, L1 = multi-position parallel prediction, L2 = block-parallel generative. L1* parallelizes frame labels; L2* denotes target self-refinement (block-parallel but not an independent drafter). "Static" indicates no runtime adaptation. ViSkip inherits the wrapped method's attributes; "group-exact" preserves the induced group variable. Rows are representative, not exhaustive.

The VLM literature thus spans L0 systems centered on visual access and candidate coverage, L1 multi-position prediction, and L2 block-parallel generative drafting. Current L2 evidence primarily comes from co-designed models in which the same backbone is explicitly trained to support block-parallel drafting and autoregressive verification. It therefore demonstrates block-level self-speculation in VLMs, while remaining distinct from an independent, low-cost parallel drafter that can be paired with heterogeneous pretrained VLM targets.

Video–language models. Video-language models amplify the condition-cost problem because long visual prefixes enlarge draft-side prefill, attention, and KV-cache costs. A small drafter may also lose accuracy when relevant evidence is diluted across the video context. Video speculative decoding has consequently focused on reducing condition-processing cost and overlapping system execution, while candidate generation remains largely autoregressive.

Sparse-to-Dense (StD) reduces draft computation without changing the input representation: it uses sparse top-K attention during drafting and restores dense attention for target verification (Zhang et al., 2025). SpecVLM instead sparsifies the condition presented to the drafter through verifier-

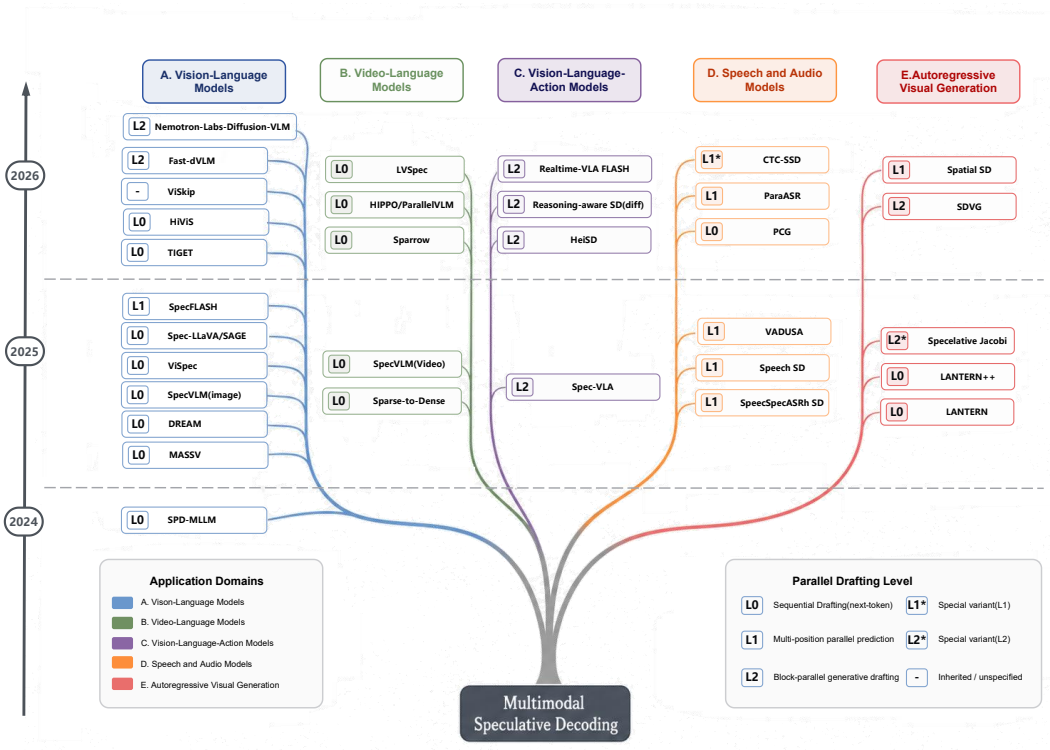


Figure 1: Taxonomy and evolution of representative multimodal speculative decoding methods.

guided and spatially uniform token pruning, while the target verifies against the full video context (Ji et al., 2025). The two systems thus apply sparsity to different objects—draft attention and draft input—while retaining strict target-token verification.

Sparrow questions whether the drafter should process video tokens at all. It attributes long-video degradation to visual KV-cache growth, context mismatch, and attention dilution, and instead reuses target-side text states that have already integrated visual information (Zhang et al., 2026b). Its recursive multi-token prediction exposes the drafter to predicted states during training; the deployed draft trajectory remains autoregressive, optionally organized as a token tree.

A separate line reduces mutual waiting between draft and target models. HIPPO combines spatial–temporal relevance pruning with an execution policy that overlaps the next proposal and current verification, adapting between optimistic and conservative schedules from recent outcomes (Lv et al., 2026). ParallelVLM similarly combines verifier-guided pruning with overlapped execution and corrects positional bias in attention-based token selection (Kong et al., 2026). This pipeline overlap is systems parallelism rather than parallel generation of future draft positions.

LVSpec modifies the acceptance contract instead. It applies strict verification to visually grounded anchors, relaxes acceptance for less visually relevant tokens, and allows nearby-position matching between target and draft sequences (Ji et al., 2026). Because this rule need not reproduce the exact target sequence, its reported preservation of more than 99.8% of target task performance is empirical high-fidelity evidence rather than a formal exactness guarantee.

Current video-specific methods are therefore concentrated at L0 but obtain acceleration from three distinct sources: condition sparsification or feature reuse, draft–verify pipeline overlap, and visually informed relaxed acceptance. Block-diffusion VLMs do not yet resolve the long-video setting, where a parallel drafter must avoid recreating the target's condition cost while preserving temporal grounding and competing with L0 systems that already prune, reuse, or hide draft computation.

Vision–language–action models. VLA systems place speculative inference inside a closed-loop control pipeline, which creates two verification problems. Under a fixed observation, exact equality between discretized action tokens may be unnecessarily strict because nearby tokens can encode functionally similar controls. Across time, however, even an initially valid action chunk can become unsafe or obsolete as new observations arrive. Verification must therefore address both candidate–target agreement and the continuing validity of an accepted plan.

Spec-VLA studies the first problem with a compact autoregressive VLA drafter and accepts a proposed action token when its discretized-bin distance from the target token is below a threshold (Wang et al., 2025a). KERV adds kinematics-aware rectification: a Kalman predictor repairs speculative errors, while a runtime signal adjusts the acceptance threshold (Zheng et al., 2026a). HeiSD instead routes geometrically irregular segments to a learned VLA drafter and regular segments to retrieved action sequences, using verify-skip and sequence-wise relaxed acceptance on the retrieval path (Zheng et al., 2026b). The learned draft paths remain autoregressive, while retrieval supplies a different candidate source. Correctness is consequently supported by task success and trajectory quality rather than exact reproduction of the target action sequence.

SV-VLA addresses the second problem explicitly. A heavy VLA acts as a low-frequency macro-planner that emits an action chunk and planning context, whereas a lightweight verifier repeatedly compares the planned action with a closed-loop reference conditioned on the latest observation and triggers replanning when needed (Wang et al., 2026b). This is not merely a looser token-matching rule: it tests whether an accepted open-loop plan remains valid after the environment and execution history have changed. Its contribution therefore lies in verification and runtime adaptation rather than in a new candidate-generation P-level.

Recent work also changes the object generated in parallel. Reasoning-aware speculative decoding accelerates the autoregressive chain of causation that precedes trajectory prediction. Its block-diffusion variant is L2 because it proposes a future reasoning block jointly, although the parallelized object is language reasoning rather than control (Dinh et al., 2026). Realtime-VLA FLASH provides direct action-side evidence: a lightweight model drafts a continuous action chunk, the target Action Expert verifies the chunk in parallel, and a phase-aware mechanism falls back to full inference when consistency is insufficient (Niu et al., 2026). We record this joint action-chunk proposal as L2, with consistency-based rather than exact verification. Fast-dDrive supplies a complementary L2 design for autonomous driving. Its block-diffusion VLA refines tokens within scaffolded semantic sections, including a trajectory section, and uses scaffold speculative decoding to accelerate the structured output (Zhang et al., 2026a).

L2 evidence within the VLA domain thus spans distinct generated objects: reasoning blocks, continuous action chunks, and structured outputs containing trajectories. The first parallelizes upstream language reasoning, whereas the latter two place speculative generation directly on action or trajectory outputs. These control-side settings must additionally validate accepted plans under evolving observations, provide safe fallback and recovery, and establish end-to-end gains across control tasks and VLA architectures.

Speech and audio models. Speech speculation spans three output settings. ASR produces a transcript that is strongly constrained by the acoustic input; audio-language systems generate more open-ended text; and speech synthesis emits long acoustic- or codec-token sequences for which different tokenizations may decode to perceptually similar waveforms. These settings offer different sources of predictability and require different notions of verification.

For ASR, SpecASR uses a smaller audio-conditioned model with adaptive draft length, sequence recycling, and sparse-tree verification (Wei et al., 2025). Its learned draft trajectory remains L0; adaptation and tree width reduce wasted work without changing its serial depth. ParaASR provides direct L1 evidence by training multiple future-transcript-token branches on an audio-language model. It proposes six transcript tokens per decoding step and commits only the prefix verified by the autoregressive head (Lin et al., 2026). CTC-based proposals parallelize a different unit. CTC-SSD reuses the acoustic encoder to predict frame labels simultaneously, collapses them into a complete transcript hypothesis, and either accepts a low-entropy hypothesis, verifies its token likelihoods in one LLM pass, or resumes autoregressive decoding from the accepted prefix (Saon et al., 2026). We denote this as L1*: it removes output-length-dependent drafting, as L1 intends, but parallelizes acoustic frames rather than directly predicting future transcript positions. SMUD uses a related CTC

preliminary transcript and batched decoder comparison to locate regions that still require causal search (Okabe & Yamamoto, 2025).

Speech synthesis shifts attention from transcript alignment to perceptual equivalence. Speech Speculative Decoding (Speech SD) employs a lightweight L0 speech LM and adds a heuristic tolerance factor to its sampling acceptance probability (Lin et al., 2025b). VADUSA instead attaches Medusa-style heads for parallel future speech-token prediction, forms sparse candidate trees, and uses tolerance-based verification; the heads make it a direct L1 speculative TTS design (Li et al., 2025a). Codec-MTP is also L1, but uses several future-token heads with Viterbi path selection rather than longest-prefix tree verification (Nguyen et al., 2025). These relaxed or path-level procedures are supported by intelligibility, naturalness, and speaker-similarity evidence, not by preservation of the target token distribution. Principled Coarse-Graining (PCG) gives this distinction a more formal treatment: it forms overlapping Acoustic Similarity Groups from the target embedding space and performs rejection sampling over the induced group distribution (Yanuka et al., 2026). Its exactness guarantee applies to the group variable, not to the target's original token sequence, so perceptual evaluation remains necessary when substituting tokens within a group.

For open-ended audio-conditioned text, UGSD drafts speech-emotion captions on an edge model and escalates only uncertain blocks to a cloud verifier, making runtime adaptation trade off latency, communication, and privacy (Xue et al., 2026).

Adjacent block-generation work provides enabling evidence for speech-side L2. Chatterbox-Flash is not itself a draft–verify speculative system: it converts an autoregressive TTS decoder into a block-diffusion decoder that generates speech tokens in parallel within streaming blocks (Seo et al., 2026). It therefore demonstrates an enabling block-generation capability rather than a complete L2 speculative pipeline. What remains unresolved is whether such a generator can serve as a sufficiently cheap proposal model for an autoregressive target, and how its blocks should be verified under an exact or perceptually justified contract without erasing the latency saved by parallel drafting.

Autoregressive visual generation. In autoregressive visual generation, the speculative object is a spatial or spatiotemporal representation, including discrete image codes, continuous latents, and video blocks. Exact identity may be unnecessarily restrictive when several codes decode to similar content, yet a local substitution can propagate through later regions. The literature therefore explores both alternative candidate-construction mechanisms and output-aware verification; only some methods parallelize proposal generation.

For discrete visual tokens, LANTERN accepts candidates over neighborhoods in the latent codebook and bounds the resulting distributional shift in total variation (Jang et al., 2025). LANTERN++ combines this relaxed contract with a static tree to avoid shallow drafts under low-confidence visual distributions (Park et al., 2025). Both retain a L0 autoregressive proposal trajectory; their main innovations lie in visual equivalence and candidate coverage rather than future-position parallelism.

Geometry-aware parallel decoding provides adjacent evidence. ZipAR is a target-side parallel decoder that predicts row- and column-direction tokens concurrently using adaptive local windows and rejection sampling analogous to speculative decoding (He et al., 2025); it is not a conventional drafter–target speculative system. Spatially Speculative Decoding (Spatial SD), by contrast, learns lightweight horizontal and vertical latent heads that draft subsequent rows in parallel, followed by target-side auto-correction (Xiang et al., 2026). These fixed-offset heads constitute L1: they predict many positions per pass without jointly modeling the entire future block. MuLo-SD supplies a complementary coarse-to-fine design, using a low-resolution drafter and restricting rejection and resampling to local neighborhoods rather than discarding the full raster-order suffix (Peruzzo et al., 2026). Beyond discrete codes, Continuous Speculative Decoding derives acceptance and rejection procedures for the diffusion distributions of continuous-valued visual AR models, with trajectory alignment improving draft–target agreement (Wang et al., 2024a). VVS instead reduces target cost by dynamically skipping selected verification rounds and reusing cached token features (Dong et al., 2026).

Speculative Jacobi Decoding (SJD) follows a drafter-free route: the target iteratively updates a future token window and commits a prefix under a probabilistic convergence criterion (Teng et al., 2025b). We annotate it as L2* because it refines a parallel block but does not use a separate low-cost drafter (the asterisk denotes block-parallel refinement without an independent drafter). GSD

inherits this SJD trajectory and replaces token-level verification with relaxed acceptance over dynamically constructed groups of visually valid codes (So et al., 2025); its contribution therefore lies primarily in the acceptance contract. SJD2 and SJD++ respectively stabilize refinement through next-clean-token prediction and high-confidence token retention (Teng et al., 2025c;a), while Speculative Coupled Decoding couples successive samples to improve stability without changing the target sampling distribution (So et al., 2026).

This application domain is also extending to video. SDVG uses a smaller denoising model to propose continuous spatiotemporal blocks for a block-autoregressive video target and verifies them with an image-quality router (Hu & Zhang, 2026). It is a L2 block-proposal design, but uses a quality-gated rather than exact distributional contract. Overall, visual generation spans L0 autoregressive drafting, geometry-aware L1 prediction, L2 block proposals, and L2* target self-refinement. Across these forms, the output-level correctness claim remains inseparable from the reported efficiency gain.

2.3 Cross-Domain Analysis

Table 2 condenses the preceding review along three cross-domain dimensions: candidate-generation parallelism, acceptance semantics, and domain-specific cues for adapting speculative computation.

Across domains, multimodal conditioning and output structure change both the cost profile of speculation and the information available for adapting it. Domain-specific cues can act on condition access, candidate construction, verification, or system execution. For example, relevance can determine which condition features are exposed, while uncertainty can govern whether speculation or verification should continue. Such adaptations can reduce condition exposure, redundant computation, or wasted verification without changing the dependence among future candidate positions. They are therefore complementary to candidate-generation parallelism rather than evidence of a higher L-level.

Acceptance contracts vary more directly with the generated object. Text-output systems can retain token-level target equivalence, whereas action, speech, and visual outputs also admit functional, sequence-level, group-level, or perceptual notions of validity. Exact contracts provide target-distribution or greedy-output equivalence under their stated assumptions. Relaxed contracts instead require empirical fidelity evidence in addition to any formal bound on their deviation from the target. This distinction becomes more consequential for block candidates, whose usefulness depends on whether they can be checked under an acceptance contract appropriate to the output space.

Across the reviewed domains, candidate generation, condition access, verification, and runtime adaptation have developed along different paths. Their effects are coupled in reported systems, leaving unresolved what is gained—and what new costs arise—when candidate generation moves from serial proposals to jointly generated or refined blocks. Addressing this question requires separating future-position dependence from domain-specific optimizations. Section 3 therefore formalizes the progression from sequential drafting to L2 block generation and examines block-parallel generative drafting within that progression. Section 4 then evaluates this mechanism empirically in multimodal inference.

Table 2: Cross-domain synthesis of multimodal speculative decoding. L-level entries report representative candidate-generation mechanisms rather than exhaustive coverage within each domain.

Application	**Domain constraint**	**Draft profile**	**Acceptance semantics**	**Domain-specific cue**
VLM	Grounding–cost trade-off	L0 / L1 / L2	Token exactness	Visual relevance
Video-LM	Long-context cost	L0	Token exactness	Temporal relevance
VLA	Closed-loop state	L0 / L2	Functional validity	Control phase
Speech/Audio	Temporal dependencies	L0 / L1	Sequence / perceptual	Prediction uncertainty
Visual Generation	Spatiotemporal coupling	L0 / L1 / L2$^{(*)}$	Group / quality	Spatial locality

Notation. L2$^{(*)}$ indicates that both block-level proposals and L2* target-side self-refinement are represented.

3 Evolution of Drafter-Side Parallelism

3.1 From Sequential to Parallel Drafting

Early speculative decoding uses a small autoregressive model to propose future tokens (Leviathan et al., 2023; Chen et al., 2023). The EAGLE series instead predicts target-model features and maps them back to token distributions (Li et al., 2024a;b; 2025b). These methods improve draft quality and can expand multiple candidate branches, but the underlying draft trajectory remains sequential: moving to a deeper future position requires another causally dependent drafter forward.

A different line of work reduces this sequential drafting depth. Medusa and multi-token prediction predict several future positions in parallel (Cai et al., 2024; Gloeckle et al., 2024). More generally, parallel drafters allow one drafter forward to advance several future positions rather than only the next one. Block-parallel generative drafting takes a further structural step. DFlash and DSpark organize the main drafting computation around an entire future block, allowing its positions to be processed in parallel (Chen et al., 2026; Cheng et al., 2026).

This progression motivates a L0–L2 taxonomy of speculative drafting. The taxonomy characterizes *drafter-side parallelism*: how far candidate generation can advance along the future sequence within one drafter forward. It is independent of how many alternative candidates are retained or how they are verified.

3.2 A L0–L2 Taxonomy of Speculative Drafting

We first define the terminology used in the taxonomy. Consider one speculation round that proposes K future positions. A *candidate trajectory* is an ordered sequence of proposed future tokens. Its *draft depth* denotes the position along this trajectory: depth 1 corresponds to the next token, depth 2 to the following token, and so on. A *drafter forward* denotes one invocation of the main drafter network. Finally, *candidate width* denotes the number of alternative candidates maintained at the same draft depth.

The L0–L2 taxonomy captures a fundamental progression: token generation $\rightarrow$ parallel position prediction $\rightarrow$ block generation. L1 parallelizes prediction across future positions; L2 changes the generative unit from individual future positions to the future block itself.

L0: Sequential drafting. L0 advances a candidate trajectory one depth at a time. To propose K consecutive future positions, the drafter therefore performs K causally dependent forwards. Small autoregressive language models and the EAGLE family belong to this regime. EAGLE may produce several alternative candidates at one depth, but these candidates increase *width*, not *depth*: reach-

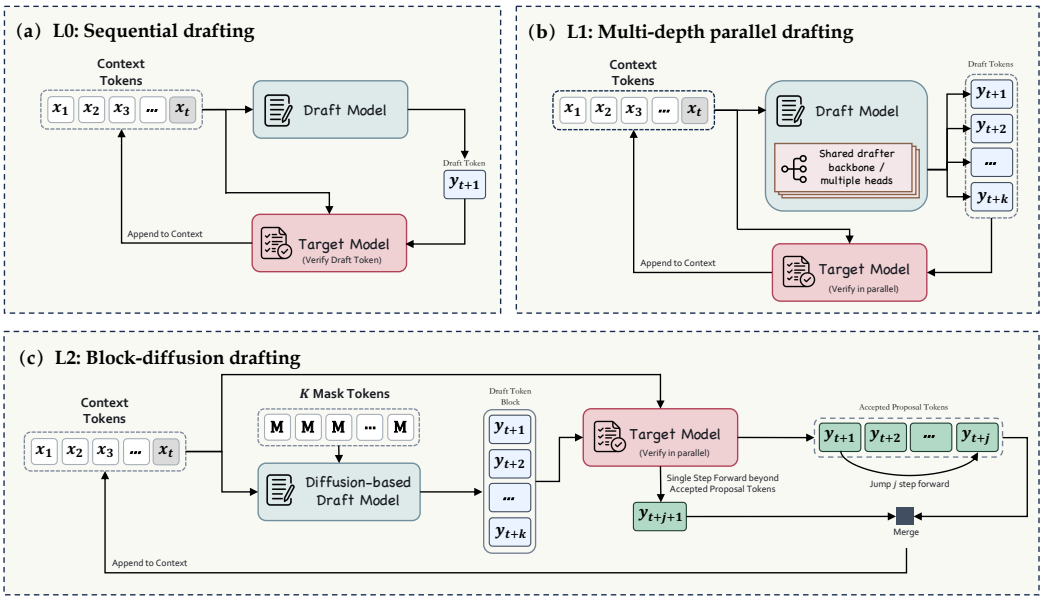


Figure 2: Illustration of the L0–L2 taxonomy of drafter-side parallelism. L0 performs sequential drafting with one future position advanced per drafter forward; L1 predicts multiple predefined future positions in parallel from a shared prefix; and L2 treats the future block as a jointly generated state, enabling block-parallel drafting followed by parallel target verification.

ing the next depth still requires another drafter forward. Mathematically, L0 produces a sequence through repeated application: $p(x_{t+1}), p(x_{t+2} \mid x_{t+1}), \ldots, p(x_{t+K} \mid x_{t:t+K-1})$.

L1: Multi-position parallel drafting. L1 breaks the one-forward–one-depth constraint by predicting several predefined future positions in parallel from shared prefix features. The definition is intentionally architecture-agnostic: parallelism may be implemented with multiple prediction heads, multi-token prediction, grouped prediction, semi-autoregressive chunks, or other parallel drafting mechanisms. What defines L1 is not the specific architecture, but that several future depths are predicted independently from a shared context rather than generated as a jointly evolving block. Mathematically, L1 produces marginals $p(x_{t+1}), p(x_{t+2}), \ldots, p(x_{t+K})$ from the accepted prefix, without joint block-level state updates.

L2: Block-parallel generative drafting. L2 identifies a block-native form of parallel drafting. Rather than treating future positions only as several predictions to be made in parallel, the drafter takes the future block itself as the basic generative unit. The entire block evolves jointly through refinement steps: $X^{(0)}_{t:t+K} \to X^{(1)}_{t:t+K} \to \cdots \to X^{(S)}_{t:t+K}$, where S is bounded and does not scale linearly with K. Block diffusion, exemplified by DFlash and DSpark (Chen et al., 2026; Cheng et al., 2026), is the representative realization considered in this paper, but L2 is not limited to diffusion architectures. Lightweight dependency modeling inside the block does not change this classification; L2 is defined by the block-parallel organization of the main drafting computation.

The distinction between L1 and L2 is therefore structural rather than a simple count of outputs per forward. Both may advance several future depths in one forward. L1 predicts several predefined future offsets from a shared prefix, whereas L2 treats the future block as a jointly evolving generative state.

Candidate width is orthogonal to draft depth. Draft parallelism should not be confused with candidate branching. Candidate width measures how many alternatives are retained at a given depth; draft depth measures how far the candidate trajectory extends into the future. Tree-based speculative decoding increases the former. SpecInfer introduces token-tree verification (Miao et al., 2024), Sequoia optimizes tree topology (Chen et al., 2024), and DDTree constructs a budgeted tree from

Table 3: A L0–L2 taxonomy of drafter-side parallelism. K denotes the number of future positions considered in one speculation round. The taxonomy describes how candidate trajectories are generated; candidate width, tree construction, and target-side verification are orthogonal.

Level	Drafting regime	Core definition	Representative methods
L0	Sequential drafting	One forward advances one draft depth; reaching depth K requires K causally dependent forwards	Small AR LM; EAGLE
L1	Multi-position parallel drafting	One forward predicts several predefined future positions in parallel from a shared prefix, but does not treat the future block as a jointly evolving generative state	Medusa; MTP; SpecFLASH
L2	Block-parallel generative drafting	The future block is the basic generative state; its positions are jointly generated or refined with bounded serial drafting depth	DFlash; DSpark

block-parallel predictions (Ringel & Romano, 2026). These mechanisms do not by themselves change the L level of the underlying drafter. An EAGLE drafter remains L0 when its candidates are organized into a tree, while DFlash combined with DDTree remains L2.

In short, L0 advances one draft depth per forward, L1 predicts multiple future positions in parallel, and L2 organizes parallel drafting around an entire future block as a jointly generative state.

3.3 The Multimodal L2 Gap

Section 2 shows that multimodal speculative decoding has advanced rapidly in condition compression, target–drafter alignment, candidate construction, and modality-aware verification. Yet most existing methods remain within L0 or L1 drafting. Evidence for L2—block-parallel generative drafting that treats the future block as the jointly evolving generative state—remains limited. In contrast, text speculative decoding already provides concrete L2 examples such as DFlash and DSpark. Text speculative decoding has entered L2; multimodal speculative decoding largely has not. We call this the *multimodal L2 gap*.

This gap does not imply that multimodal models are inherently unsuitable for block-parallel generative drafting. Multimodal conditioning instead changes the tradeoff. Accessing visual, audio, or other modality features can increase draft cost, while the additional condition may also make future tokens more predictable. A useful multimodal L2 drafter must therefore exploit the condition well enough to produce longer or higher-quality candidate blocks while preserving the efficiency of block-parallel drafting.

This motivates the controlled experiments in Section 4. Following prior speculative-decoding surveys (Xia et al., 2024; Hu et al., 2025a), we evaluate both draft quality and system efficiency. We report average accepted length, token acceptance rate, target-token rank, and full-block acceptance, together with stage latency, throughput, decode-only and end-to-end speedup, memory, and energy where available.

4 Is Multimodal Speculative Decoding Ready for L2?

4.1 Evaluation Questions and Experimental Protocol

Sections 2 and 3 show that multimodal speculative decoding has made substantial progress in condition compression, candidate construction, and verification strategies, while the transition from sequential or multi-position drafting to L2 block-parallel generative drafting remains unclear. Unlike text-only models, multimodal systems introduce additional factors, including heterogeneous archi-

tectures, expensive modality conditioning, task-dependent predictability, and system-level overhead. Therefore, we study L2 readiness through a diagnostic perspective: rather than asking whether a block-parallel drafter can be constructed, we ask under which model, task, condition, and system configurations it can translate draft quality improvements into practical end-to-end acceleration.

We organize our empirical study around four evaluation questions:

Q1: Model compatibility. Can existing multimodal architectures support effective L2 block-parallel drafting? We evaluate whether lightweight L2 drafters can achieve sufficient draft–target agreement across different multimodal model families, scales, and training paradigms. This analysis examines whether L2 readiness is a general property of multimodal models or depends on specific target architectures and drafter training strategies.

Q2: Conditioning efficiency. Does multimodal conditioning remain the dominant bottleneck after draft generation becomes parallel? Although L2 drafting substantially reduces the cost of generating future tokens, multimodal drafters may require additional computation to access visual or other modality information. We analyze the latency contribution of conditioning, and investigate whether reducing or removing explicit modality access can preserve draft quality while improving end-to-end efficiency.

Q3: Task and input dependence. When do improved draft quality and accepted length translate into practical acceleration? Multimodal generation workloads differ significantly in output uncertainty, grounding requirements, and input complexity. We therefore evaluate L2 drafting across diverse tasks and input conditions to identify when the accepted-token gain is sufficient to amortize additional speculative overheads.

Q4: System readiness. Is the current inference ecosystem mature enough to support practical multimodal L2 deployment? Beyond algorithmic effectiveness, practical adoption requires integrated support for drafter training, candidate generation, verification, and serving frameworks. We summarize the current ecosystem support for representative L2 methods and discuss remaining deployment barriers.

Based on these questions, we conduct a cross-model evaluation covering different multimodal architectures and task categories. We report both draft quality and system efficiency metrics, including mean accepted tokens (MAT), token acceptance rate, target-token rank, block acceptance, stage-level latency, throughput, memory consumption, and end-to-end speedup. Unless otherwise specified, all experiments use matched decoding configurations and backend-specific autoregressive baselines to ensure fair comparison. We use a common protocol to evaluate the L2 transition across model families and tasks. Numerical cells are deliberately marked "TBD" until the final reproducible export; we do not infer or estimate missing measurements.

4.2 Experimental Setup

Models and implementation. Our evaluation suite comprises four vision-language models selected to span diverse configurations along three orthogonal axes: model scale, architectural family, and pretraining regime. Concretely, we include Qwen3-VL-4B and Qwen3-VL-8B (Bai et al., 2025), which are natively pretrained as multimodal dense models; Qwen3.6-27B (Qwen Team, 2026a), a dense model initialized from text-only pretraining and subsequently aligned to multimodal inputs; and Qwen3.6-35A3B (Qwen Team, 2026b), a Mixture-of-Experts (MoE) model that likewise undergoes text pretraining followed by multimodal alignment. This selection spans parameter counts from 4B to 27B (active parameters for MoE) and covers both end-to-end multimodal pretraining and post-hoc multimodal alignment, mitigating the risk that conclusions are confounded by a single architectural choice or training recipe. For targets with publicly released speculative decoding checkpoints—most notably Qwen3.6-27B—we compare up to five decoding configurations: (1) target-only autoregressive (AR) decoding, which serves as the speed and quality baseline; (2) EAGLE-3 (Li et al., 2025b) with the PRISM-EAGLE3 drafter checkpoint released by Ex0bit (L0); (3) the native multi-token prediction (MTP) heads (Gloeckle et al., 2024) shipped with Qwen3.6 (L1); (4) DFlash (Chen et al., 2026) with the publicly released z-lab checkpoint (L2); and (5)

https://huggingface.co/Ex0bit/Qwen3.6-27B-PRISM-EAGLE3
https://huggingface.co/z-lab/Qwen3.6-27B-DFlash

DSpark (Cheng et al., 2026) with the publicly released satgeze checkpoint (L2). For targets without publicly available L2 checkpoints—in particular Qwen3-VL-8B—we additionally train a DFlash-style block-parallel drafter using the SpecForge training framework (Li et al., 2026), ensuring that our evaluation is not limited to targets for which pre-trained drafters happen to be publicly released.

Unless otherwise noted (see † in Table 4), all experiments are conducted on the SGLang backend under matched greedy decoding settings, with a maximum generation length of 2,048 new tokens and a fixed random seed of 42. All methods employ identical prompts, tokenizers, stopping criteria, and batch sizes. We report end-to-end wall-clock speedup, which encompasses vision encoding, prefilling, drafter candidate generation, tree construction, and target model verification.

Tasks and evaluation matrix. We construct a 600-sample evaluation set following the task balance of MMSpec (Shen et al., 2026), a dedicated benchmark for multimodal speculative decoding. Concretely, we randomly sample 100 instances from each of six task categories using a fixed random seed of 42: general VQA from GQA (Ainslie et al., 2023), image captioning from Flickr30K (Plummer et al., 2015), text-focused VQA from TextVQA (Singh et al., 2019), chart understanding from CharXiv (Wang et al., 2024b), complex multimodal reasoning from MMMU (Yue et al., 2024), and multi-turn conversation from the combined ConvBench (Liu et al., 2024) and MM-MT-Bench (Agrawal et al., 2024) splits. The only deviation from MMSpec is that we replace the COCO captioning split with Flickr30K to reduce lexical overlap with common VLM pre-training corpora.

Baselines and metrics. The primary baseline is target-only autoregressive decoding executed within the same inference backend and under identical decoding settings. Candidate quality is measured by mean accepted tokens (MAT), defined as the average number of consecutive draft tokens accepted by the target per speculation round. The primary efficiency metric is end-to-end wall-clock speedup, which encompasses vision encoding, target prefill, drafter prefill, candidate generation, tree construction, target verification, and token sampling. Speedup is computed against the matched target-only baseline:

$$\text{Speedup} = \frac{T_{\text{AR}}}{T_{\text{method}}}. \tag{2}$$

The experiments that follow answer the four questions through three empirical analyses: cross-model effectiveness evaluation, conditioning and input-scaling analysis, and ecosystem support analysis.

4.3 Does L2 Block-Parallel Drafting Deliver Multimodal Speedups?

We begin with the foundational effectiveness question: does L2 block-parallel drafting brings the acceptance-length and latency gains observed in text-only speculative decoding to multimodal autoregressive models, or does the additional draft-side computation and multimodal conditioning overhead negate these benefits? To guard against conclusions that are artifacts of a single architecture, we evaluate across the four-model suite described above, which jointly varies model scale (4B–27B total parameters), architectural family (dense vs. MoE), and pretraining regime (native multimodal vs. text-pretrained then multimodal aligned). For Qwen3-VL-8B, where no public L2 drafter checkpoint is available, we train a DFlash-style block-parallel drafter via SpecForge (Li et al., 2026) to avoid selection bias toward targets with pre-existing third-party checkpoints. We complement these head-to-head comparisons with a general bench collected in 4.2 —to identify the specific conditions under which L2 drafting succeeds or fails.

Table 4 extends the Qwen3.6-27B comparison to the Qwen3-VL family. On Qwen3.6-27B, under a matched SGLang backend, evaluation examples, and decoding protocol, DFlash achieves the highest MAT and speedup on every benchmark, with a task-equal MAT of 4.38 and an average speedup of 2.60× over AR; DSpark ranks second with 3.52 MAT and 2.04× speedup. On the matched HuggingFace (HF) subsets†, DFlash reaches 2.50 MAT and 1.95× speedup for Qwen3-VL-4B, and 2.75 MAT and 2.14× speedup for Qwen3-VL-8B. By contrast, the SGLang EAGLE-3 runs remain below their matched SGLang AR baselines, at 0.71× and 0.88× overall, respectively. Because the

https://huggingface.co/satgeze/Qwen3.6-27B-DSpark

Table 4: Speculative decoding performance with a maximum of 2,048 new tokens. MAT denotes the mean number of output tokens advanced per speculative step. Overall results are task-equal means. The best and second-best speculative results for each target are shown in bold and underlined, respectively. †Results measured on the HuggingFace Transformers backend; all other results use the SGLang backend. Each speedup is normalized to its own backend-matched autoregressive baseline. We abbreviate target models Q denotes Qwen.zheli

Target	Method	GQA		F30K		TextVQA		CharXiv		MMMU		MT		Overall	
		MAT	Speedup	MAT	Speedup	MAT	Speedup	MAT	Speedup	MAT	Speedup	MAT	Speedup	MAT	Speedup
Q3VL-4B	AR	–	1.00×	–	1.00×	–	1.00×	–	1.00×	–	1.00×	–	1.00×	–	1.00×
	EAGLE-3	1.95	0.71×	1.96	0.67×	1.98	0.71×	2.28	0.76×	**2.41**	0.81×	**2.14**	0.60×	2.12	0.71×
	DFlash†	**3.05**	**2.28**×	**2.84**	**2.36**×	**2.05**	**1.57**×	**2.63**	**2.06**×	2.28	**1.74**×	2.13	**1.70**×	**2.50**	**1.95**×
Q3VL-8B	AR	–	1.00×	–	1.00×	–	1.00×	–	1.00×	–	1.00×	–	1.00×	–	1.00×
	EAGLE-3	2.22	0.85×	2.20	0.85×	2.05	0.81×	2.66	0.99×	2.48	0.95×	**2.17**	0.85×	2.30	0.88×
	DFlash†	**2.45**	**1.78**×	**2.89**	**2.18**×	**2.53**	**1.87**×	**3.45**	**2.73**×	**3.09**	**2.55**×	2.08	**1.72**×	**2.75**	**2.14**×
Q3.6-27B	AR	–	1.00×	–	1.00×	–	1.00×	–	1.00×	–	1.00×	–	1.00×	–	1.00×
	EAGLE-3	2.37	1.74×	2.19	1.64×	2.38	1.70×	2.50	1.83×	2.23	1.64×	2.20	1.62×	2.31	1.69×
	MTP	3.22	1.64×	3.16	1.63×	3.26	1.64×	3.39	1.72×	3.32	1.71×	2.99	1.50×	3.22	1.64×
	DFlash	**4.19**	**2.42**×	**4.30**	**2.12**×	**4.16**	**2.53**×	**5.12**	**3.19**×	**4.35**	**3.01**×	**4.17**	**2.30**×	**4.38**	**2.60**×
	DSpark	3.27	1.92×	3.26	1.69×	3.36	1.98×	4.42	2.53×	3.44	2.34×	3.35	1.76×	3.52	2.04×

Qwen3-VL EAGLE-3 and DFlash measurements use different backends, their absolute latency and throughput values are not compared directly; each reported speedup is normalized only to its own backend-matched AR run.

Four observations stand out.

First, drafter capacity appears to raise the attainable acceptance ceiling: DFlash uses approximately 1.73B parameters, compared with approximately 0.43B for Qwen3.6's native MTP module and 0.60B for EAGLE-3, and produces substantially longer accepted sequences. However, EAGLE-3 does not outperform the smaller MTP module, showing that scale alone is insufficient and that drafting architecture and training objective also matter.

Second, checkpoint provenance matters. Despite using a larger drafter, the third-party DSpark checkpoint trails DFlash on all six multimodal tasks. This gap contrasts with the strong text-only results reported in DSpark (Cheng et al., 2026), suggesting that text performance does not reliably predict multimodal acceptance quality.

Third, natively multimodal-trained models better support speculative decoding. Qwen3.6-27B, which is natively trained with multimodal data, consistently benefits from speculative decoding across different methods, achieving $1.69\times$ with EAGLE-3, $2.04\times$ with DSpark, and $2.60\times$ with DFlash on the SGLang backend. In contrast, on the Qwen3-VL family, EAGLE-3 fails to provide speedup on SGLang ($0.71\times$ for 4B and $0.88\times$ for 8B). Even when we explicitly train a DFlash-style block-parallel drafter for Qwen3-VL-8B with multimodal data using SpecForge, it reaches only $2.14\times$ speedup on HF, below the $2.60\times$ achieved by the off-the-shelf z-lab DFlash checkpoint on Qwen3.6-27B. These results suggest that native multimodal training may induce a more unified feature distribution and better alignment between multimodal representations and future-token prediction, making the target model inherently easier for speculative drafters to approximate than adding multimodal signals only during post-hoc drafter training.

Fourth, larger target models benefit more from speculative decoding. We observe a clear improvement as the target model grows. For DFlash, speedup increases from $1.95\times$ on Qwen3-VL-4B to $2.14\times$ on Qwen3-VL-8B, and reaches $2.60\times$ on Qwen3.6-27B. EAGLE-3 shows an even sharper transition, from no net acceleration on 4B ($0.71\times$) and 8B ($0.88\times$) to $1.69\times$ on 27B. Two factors likely contribute. First, DFlash uses a shallow, roughly fixed-cost 5-layer drafter, while the target-model forward cost grows substantially with model size; the drafting overhead is therefore increasingly amortized by each accepted block. Second, larger targets also appear easier to speculate: DFlash MAT rises from 2.50 and 2.75 on 4B and 8B to 4.38 on 27B. This suggests that larger

models may expose more stable and predictable feature distributions, allowing the drafter to better match future-token representations. Larger targets thus benefit from both a more favorable compute ratio and higher draft predictability.

4.4 Does Multimodal Conditioning Remain a Bottleneck Under Block Parallelism?

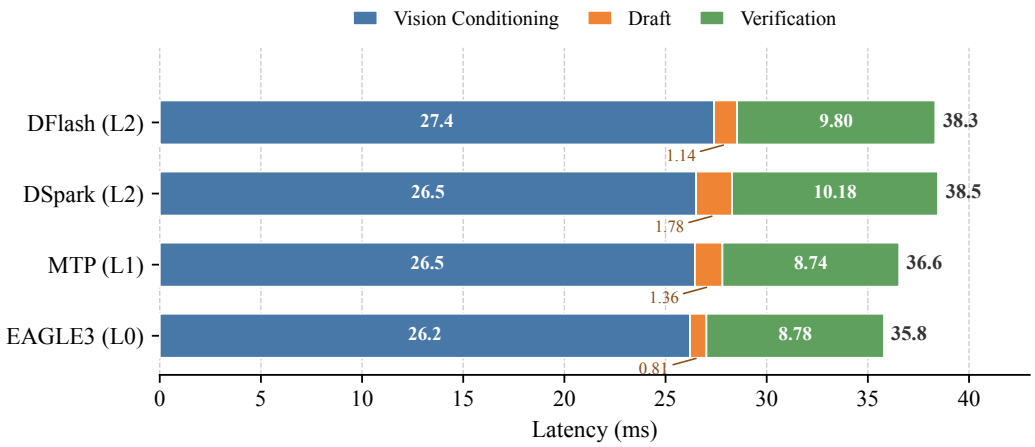


Figure 3: Stage-level latency decomposition of a representative decoding pass on Qwen3.6-27B. Vision conditioning includes vision encoding, multimodal prefill, and drafter-side target-feature-to-KV construction when applicable.

Table 5: HR-Bench results under 4K and 8K resolutions on Qwen3-VL, with a 512-token generation cap on a common 30-sample manifest. MAT denotes the mean number of output tokens advanced per speculative step. Speedup is end-to-end latency speedup over the backend-matched autoregressive baseline. Top1 denotes top-1 acceptance; Sampling denotes speculate sampling.

Resolution	Qwen3-VL-4B				Qwen3-VL-8B			
	Top1		Sampling		Top1		Sampling	
	MAT	Speedup	MAT	Speedup	MAT	Speedup	MAT	Speedup
HR-Bench 4K	2.90	1.03×	2.45	0.96×	3.22	1.06×	2.88	1.11×
HR-Bench 8K	2.86	0.90×	2.70	0.83×	3.09	0.99×	3.22	0.85×

Multimodal prefill, not block drafting, bounds end-to-end speedup. A central tension in multimodal speculative decoding is the cost of accessing the visual condition. A condition-light drafter is cheap but poorly aligned with visually grounded continuations; a fully conditioned drafter improves alignment but may duplicate part of the target-side multimodal prefill. Block-parallel drafting changes this trade-off, but does not remove it.

Figure 3 makes this clear. For L2 drafters, draft generation itself is no longer the bottleneck: DFlash spends only 1.14 ms on drafting and DSpark 1.78 ms. This confirms the main benefit of block parallelism: proposing a longer block is cheap. However, the dominant term is still multimodal conditioning. MTP, DSpark, and DFlash spend 26.45–27.40 ms in the vision-conditioning stage, far larger than the draft step and even larger than target verification. DFlash is especially informative here. Its drafter is fast, but it conditions on target hidden states by projecting them into the drafter KV cache. This target-feature-to-KV construction improves draft quality, yet it also introduces a prefill-like cost that is tied to the visual sequence length. Thus, DFlash removes most of the token-drafting cost, but not the cost of preparing the multimodal condition.

We further stress this effect on HR-Bench, using Qwen3-VL at 4K and 8K resolutions. Table 5 shows that the drafter still obtains non-trivial MAT, staying around 2.5–3.2 across models and ver-

ification rules. However, this does not translate into end-to-end acceleration. At 4K, speedups are already marginal, ranging from $0.96\times$ to $1.11\times$. At 8K, all settings fall to or below the autoregressive baseline, with speedups of $0.90\times$ and $0.83\times$ on Qwen3-VL-4B, and $0.99\times$ and $0.85\times$ on Qwen3-VL-8B. The contrast is especially clear under rejection sampling on Qwen3-VL-8B: MAT increases from 2.88 to 3.22 when moving from 4K to 8K, yet speedup drops from $1.11\times$ to $0.85\times$. Thus, the failure mode is not simply poor draft quality. Rather, higher-resolution inputs increase visual prefill and drafter-side cache-construction cost, while MAT does not grow enough to amortize this added conditioning overhead.

These two measurements point to the same conclusion. Under block-parallel speculative decoding, drafting one more token is almost free; conditioning on one more visual token is not. The end-to-end speedup is therefore bounded by multimodal prefill, especially for DFlash-style drafters that additionally convert target features into drafter-side KV cache. Future gains should thus come less from further accelerating the draft forward pass, and more from reducing or reusing the conditioning path: compressed visual tokens, target-feature reuse, shared drafter caches, or lightweight visual-state adapters.

Đúng những cái mình đã làm

Acceptance is robust to context removal. Table 6 further clarifies what kind of bottleneck multimodal conditioning creates under block parallelism. DFlash acceptance is only weakly sensitive to explicit visual conditioning in the drafter context: masking visual-token K/Vs reduces Overall MAT by only 4.2% on Qwen3-VL-4B and 3.9% on Qwen3-VL-8B. Even removing the entire original prefill context lowers MAT by only 14.1% and 11.0%, respectively. Since the target model always receives the full multimodal input, this ablation isolates the drafter-side source of acceptance. The results suggest that the diffusion-based drafter does not exploit visual evidence in the same way as a full VLM target during autoregressive generation. Instead, a large fraction of accepted draft tokens can be predicted from local continuation statistics and recent contextual states, rather than from detailed re-access to the full question-image input.

The degradation is also task dependent rather than uniform. TextVQA is the most sensitive to context removal, especially on Qwen3-VL-8B (2.750 to 1.953 under no-prefill context), whereas MMMU remains comparatively stable and GQA shows no consistent monotonic drop. This pattern suggests that explicit source conditioning matters most when the next-token distribution is tightly constrained by localized visual or OCR evidence, but contributes less when acceptance is dominated by short-range target-token continuation. A larger target also appears slightly more robust to drafter-side context ablation, which may indicate that stronger models expose more stable local continuation patterns for speculative matching.

4.5 Training and Inference Framework Support

Beyond algorithmic effectiveness, the practical adoption of L2 drafting also depends on the availability of training and inference infrastructure. Block-parallel drafting requires dedicated support for drafter training, block-level proposal generation, and speculative verification. We therefore summarize the current open-source ecosystem for representative L2 methods, focusing on both training frameworks and inference backends.

As summarized in Table 7, support for L2 drafting has expanded rapidly beyond method-specific research implementations. The official DFlash repository provides released checkpoints across multiple model families, including Qwen3, Qwen3.5/3.6, and Gemma, together with inference support through Transformers, MLX, SGLang, and vLLM. DeepSpec provides a unified training and evaluation pipeline for DFlash and DSpark on Qwen3 and Gemma models. AngelSpec supports several block-parallel architectures, including DFly, DFlash, DFlare, and DSpark, within a unified training framework using vLLM, SGLang, or HuggingFace-based inference backends. Speculators provides DFlash and DSpark training with direct integration into the vLLM ecosystem, while SpecForge provides corresponding support for DFlash, Domino, and DSpark with direct integration into SGLang. The model families listed in the table indicate representative documented configurations rather than exhaustive model–method combinations.

At the inference level, vLLM and SGLang provide the most complete production-oriented support for block-parallel speculative decoding. Both frameworks support DFlash-style drafting, and support for newer L2 methods such as DSpark is also being integrated into their speculative-decoding

Table 6: Effect of multimodal target context on DFlash speculative acceptance. Results use top-1 verification with 100 examples per task and a maximum generation length of 2,048 tokens. Overall MAT is the equal-weight arithmetic mean over GQA, TextVQA, and MMMU. Text-only context masks visual-token K/V entries in the DFlash context, whereas no-prefill context masks the complete original prefill context. The target model always receives the full multimodal input.

Model	DFlash context setting	GQA	TextVQA	MMMU	Overall MAT ↑	Δ Overall vs. full
Qwen3-VL-8B	Full multimodal context	2.692	2.750	3.259	**2.900**	–
	Text-only context	2.608	2.504	3.246	2.786	-0.114 (-3.9%)
	No-prefill context	2.845	1.953	2.946	2.581	-0.319 (-11.0%)
Qwen3-VL-4B	Full multimodal context	2.587	2.305	2.583	**2.491**	–
	Text-only context	2.487	2.221	2.453	2.387	-0.104 (-4.2%)
	No-prefill context	2.095	1.904	2.417	2.139	-0.353 (-14.1%)

Table 7: Representative framework support for L2 speculative decoding.

Repository	**Models**	**L2 Methods**	**Framework Support**
DFlash	Qwen3/3.5/3.6, Gemma4	DFlash	MLX, SGLang, vLLM
DeepSpec	Qwen3, Gemma4	DFlash, DSpark	Train, Eval
AngelSpec	Qwen3, Hunyuan3	DFly, DFlash, DFlare, DSpark	Train, vLLM, SGLang
Speculators	Qwen3/3.6, Gemma4	DFlash, DSpark	Train, vLLM
SpecForge	Qwen3/3.6	DFlash, Domino, DSpark	Train, SGLang

stacks. Transformers provides a more general assisted-decoding interface and has begun to incorporate model-specific DFlash support. MLX provides an additional execution path for DFlash on Apple Silicon, although its L2 coverage is currently narrower than that of vLLM and SGLang.

Overall, L2 speculative decoding is transitioning from isolated research implementations toward integrated training–serving ecosystems. In particular, the Speculators–vLLM and SpecForge–SGLang stacks provide increasingly complete paths from drafter training to efficient inference. However, support remains uneven across model families and is still substantially less mature for multimodal targets than for text-only models. This ecosystem gap constitutes an additional practical barrier to the broader adoption of multimodal L2 drafting.

5 L2 Readiness and Future Directions

5.1 What Determines L2 Readiness?

Taken together, the results in Section 4 suggest that L2 readiness should not be treated as a fixed property of an entire modality or model family. Instead, it depends on the specific target model, multimodal condition, task and input regime, and inference environment. We therefore interpret readiness at the model–task–condition–system level.

Model compatibility. L2 drafting is effective across the evaluated targets, but its benefit varies substantially across model families. DFlash achieves positive end-to-end speedups on both Qwen3-VL-4B and Qwen3-VL-8B, while obtaining substantially higher MAT and speedup on Qwen3.6-27B. These results indicate that multimodal L2 drafting is feasible, but not uniformly effective across targets. Importantly, the observed differences should not yet be attributed to a single architectural factor, since model scale, drafter checkpoint, training recipe, and inference backend also vary across configurations. The current evidence therefore supports *model-dependent compatibility* rather than an architecture-agnostic notion of L2 readiness.

Multimodal conditioning. The strongest multimodal-specific bottleneck is the cost of conditioning. The latency decomposition in Figure 3 shows that block-parallel draft generation itself is not the dominant cost in our evaluated configurations, whereas multimodal conditioning contributes a substantial portion of the speculative decoding overhead. Our conditioning analysis further suggests

that repeatedly exposing the drafter to the full multimodal context is not always necessary for maintaining useful speculative acceptance. A L2-ready multimodal system should therefore provide a lightweight conditioning path that preserves the information needed for drafting without reproducing the target model's multimodal processing cost.

Task and input dependence. Readiness also varies with the workload. Table 4 shows noticeable task-level variation in MAT and end-to-end speedup, indicating that the predictability of future tokens differs across captioning, VQA, OCR, and reasoning tasks. More importantly, the HR-Bench results show that relatively high MAT does not guarantee acceleration under expensive visual inputs: increasing the image resolution from 4K to 8K preserves MAT at roughly the same level while reducing end-to-end speedup to around or below the autoregressive baseline. Thus, L2 readiness depends not only on draft quality, but also on whether the accepted-token gain is sufficient to amortize the multimodal input cost. Dynamic routing across P levels or block sizes is a promising consequence of this observation, although we leave generation-stage routing to future work.

System and ecosystem support. Practical readiness further depends on whether L2 methods can be trained and deployed using existing infrastructure. As summarized in Table 7, DFlash and DSpark are increasingly supported by dedicated training frameworks such as DeepSpec, AngelSpec, Speculators, and SpecForge, while vLLM and SGLang provide increasingly mature serving paths for block-parallel speculative decoding. Transformers and MLX also provide more limited L2 execution paths. This trend shows that L2 is moving beyond isolated research implementations toward integrated training–serving ecosystems. However, support remains substantially less uniform for multimodal targets than for text-only models.

Overall readiness. The evidence therefore supports a qualified conclusion: multimodal speculative decoding is *partially ready* for L2 block-parallel drafting. L2 can already provide substantial gains for favorable target models and workloads, demonstrating that multimodality itself is not a fundamental barrier. However, these gains remain sensitive to target-model compatibility, multimodal conditioning cost, input characteristics, and framework support. The practical question is therefore not whether multimodal L2 works in general, but under which model–task–condition–system configurations its accepted-token gains are large enough to translate into end-to-end acceleration.

5.2 Challenges and Opportunities

Lightweight multimodal condition injection. Future drafters should learn the smallest multimodal representation sufficient for next-block prediction. Promising routes include target-produced semantic summaries, query-based visual compression, layer-adaptive feature selection, and dynamic routing that enables visual conditioning only when token uncertainty indicates that it is needed. The objective should jointly optimize condition sufficiency and added prefill latency.

Dependency modeling for L2 drafting. L2 drafters need enough block-internal structure to preserve consistency without returning to K serial steps. Lightweight causal correction, confidence-ordered refinement, local remasking, and cross-position state transfer may offer a better latency–quality balance than either fully independent or fully autoregressive drafting.

Dynamic Parallel levels and block sizes. No single drafter is likely to dominate all multimodal tasks. A practical system can route between an autoregressive drafter, multi-token heads (L1), DFlash (L2), DSpark (L2), retrieval, and tree expansion according to task type, current entropy, generation stage, and measured break-even cost. When the predicted gain falls below zero, the system should shorten the block or revert to target autoregressive decoding.

Tree-aware parallel-drafter training. Position-wise imitation does not directly optimize the accepted prefix or the utility of a finite verification tree. In block-parallel speculative decoding methods like DFlash, where the drafter is heavily conditioned on target model representations (via multi-layer hidden state injection), the target's top-1 token is frequently covered within the drafter's top-k candidates. For instance, the top-6 draft tokens can already cover 98–99% of the target probability mass, making exact position-wise matching largely unnecessary. Consequently, training objectives should instead reward recoverable target-consistent paths, calibrate marginal probabilities for budgeted tree

construction, and concentrate capacity on frontier positions that limit acceptance. This bridges multimodal L2 drafting with recent tree-based verification methods, rather than treating the drafter and verifier as independent modules.

Multimodal verification and relaxed acceptance. Exact token verification is appropriate for lossless text generation, but actions, codec tokens, and visual codebooks may admit functionally or perceptually equivalent outputs. Future systems should define output-space-specific acceptance rules together with explicit quality guarantees, and should distinguish relaxed verification from merely skipping target computation.

Standardized evaluation and system co-design. Parallel drafting changes the workload presented to the target: longer verification sequences and wider trees can improve arithmetic intensity but also increase memory traffic and batch interference. A shared benchmark should span architectures, modalities, input lengths, task uncertainty, output lengths, batch sizes, hardware, and backends. It should report candidate and task quality, stage latency, throughput, memory, energy, and both decode-only and end-to-end speedup. Efficient tree attention, KV-cache reuse, overlapped execution, and scheduling should be evaluated under the same protocol so that readiness becomes a reproducible model–task–system measurement.

6 CONCLUSION

This paper asks whether multimodal speculative decoding is ready to move from L0/L1 drafting to L2 block-parallel generative drafting. Our modality-centered review shows that VLM, Video-LM, VLA, Speech/Audio, and Visual-AR research has already developed strong mechanisms for condition access, token compression, candidate construction, and task-aware verification. However, systematic evidence for L2 block-parallel drafting across multimodal architectures remains limited. Our cross-architecture study directly examines this transition and reveals a more nuanced picture: neither a universal failure nor a universal success. While direct L2 transfer can be weak on some multimodal architectures, newer architectures demonstrate substantially stronger draft–target compatibility. These findings suggest that multimodal L2 readiness is configuration-dependent, determined by architectural compatibility, condition accessibility, task predictability, dependency handling, and system-level amortization. Therefore, multimodal speculative decoding is partially ready for diffusion-based parallel drafting, but practical deployment still requires careful co-design of the model architecture, conditioning pathway, and inference system.

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